# BHOOMI RAKSHAK (■■■■ ■■■■■)

## Deep-Dive Technical Stack & System Architecture Specification

**Problem Statement:** SIH 2026 #26191 (Disaster Risk & Relocation Planning Platform)
**Document Type:** Comprehensive Engineering Reference & Technical Blueprint
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## 1. Executive Summary & Architectural Philosophy

**Bhoomi Rakshak** is an enterprise disaster-risk mitigation, habitational vulnerability assessment, and proactive relocation planning system designed for District Disaster Management Authorities (DDMAs) and State Disaster Management Authorities (SDMAs). The platform combines deterministic geospatial hazard overlay algorithms, multi-modal 2D/3D GIS visualization, carrying-capacity suitability auditing, and GenAI advisory assistance.

The system employs a **Decoupled Dual-Mode Architecture**: (1) A standalone, zero-latency client-side simulation capable of rendering and computing over 1,000+ habitational red-zones entirely within the browser using WebAssembly and client-side graph algorithms; (2) An asynchronous RESTful spatial engine powered by Python FastAPI, GeoPandas, and PostGIS for heavy spatial intersections, real-time telemetry fusion, and automated legal certificate rendering.

| Tier | Primary Technologies | Key Functional Responsibilities |
|---|---|---|
| **Presentation & GIS Client** | React 19, Vite 8, React-Leaflet 5, Leaflet 1.9, Three.js, Recharts, Framer Motion | Multi-basemap satellite GIS, 3D raycasted geological terrain, Dijkstra corridor routing, bilingual voice advisories, client-side caching. |
| **Application & Spatial API** | FastAPI 0.141, Uvicorn ASGI, Pydantic v2, GeoPandas 1.1, Shapely 2.1, PyProj | Multi-hazard overlay analysis, vulnerability weighting, live USGS seismic ingestion, model retraining, and legal certificate compilation. |
| **Data & Persistence** | PostgreSQL 15 + PostGIS 3.3, SQLite 3 (Dev), SQLAlchemy 2.0 ORM, Asyncpg | Spatial boundary indexing (R-Tree), habitational census attributes, candidate relocation parcel metrics, citizen observation logs. |
| **AI & Integrations** | Google Gemini 1.5/Flash API, USGS Seismic API, Ethereal SMTP Mailer | Natural language civic disaster assistant, real-time seismic event streaming, automated dispatch alerting. |

## 2. Frontend Layer (Client-Side Engineering)

Built on **React 19** and bundled with **Vite 8**, the client application achieves instantaneous load speeds and 60fps mapping performance. The interface includes deep-linkable administrative workspaces (/console), public citizen safety portals (/citizen), and executive dashboards.

| Component | Library / Spec | Technical Purpose & Architecture |
|---|---|---|
| **Core Framework** | React 19.2.8 + React DOM | Exploits concurrent rendering, fine-grained state management, and optimized virtual DOM diffing across 1,000+ simulated spatial points. |
| **Build Pipeline** | Vite 8.2.2 + Rollup | ESM-based HMR dev server; production bundle code-split with React.lazy and Suspense for under-200ms initial interactive load. |
| **Routing Engine** | React Router DOM 7.18.3 | Declarative client-side routing with URL deep-linking for zone parameters (/console?zone=BR-104) and 404 fallback routing. |
| **2D GIS Mapping** | Leaflet 1.9.4 + React-Leaflet 5.0 | Multi-layer basemap switching (Esri High-Resolution World Imagery, OSM Streets, OpenTopoMap). Custom SVG vector pins, animated flight bounds. |
| **3D Geological Engine** | Three.js 0.185.1 (WebGL) | Procedural 3D elevation mesh with fault line extrusion, contour raycasting, dynamic lighting, and orbital camera focus on active zones. |
| **Data Analytics Charts** | Recharts 3.10.1 | SVG-based responsive visualizations: 8-factor radar suitability charts, priority score distributions, and carrying capacity shortfall bars. |
| **Animation & Icons** | Framer Motion 13.3 + Lucide React | Hardware-accelerated layout springs, scenario shift transitions, and 200+ accessible SVG vector icons. |

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|---|---|---|
| **Speech & Geolocation** | Web Speech API + HTML5 GPS | Bilingual (English & Hindi) client-side text-to-speech advisory player; real-time GPS coordinate triangulation snapping to nearest hazards. |

## 3. Backend Layer & Geospatial Computation Engine

The server-side application is developed in **Python 3.13** utilizing **FastAPI** for asynchronous request dispatching and **GeoPandas / Shapely** for rigorous vector geometry intersections and spatial overlays.

| Module | Technology / Library | Engineering Functionality |
|---|---|---|
| **RESTful Framework** | FastAPI 0.141.1 + Starlette | Asynchronous ASGI endpoints with auto-generated OpenAPI / Swagger documentation (/docs). |
| **ASGI Web Server** | Uvicorn 0.52.4 | Production-grade asynchronous web server supporting connection concurrency, keep-alives, and multi-worker pooling. |
| **Spatial Joins & Overlays** | GeoPandas 1.1.4 + Shapely 2.1.2 | Performs vector intersections between dynamic hazard polygons and habitational centroids using gpd.overlay(how='intersection'). |
| **Cartographic Projections** | PyProj 3.8.0 + PyOgrio 0.13.0 | Handles geodesic distance calculations and coordinate transformations (WGS84 EPSG:4326 to UTM projection systems). |
| **Statistical Processing** | Pandas 3.0.5 + NumPy 2.5.3 | Vectorized demographic aggregation (elderly %, disabled %, housing structural index) and score normalization matrices. |
| **Dynamic Model Retraining** | Scikit-Learn 1.9 + SciPy 1.18 | Continuous model calibration endpoint (/api/model/retrain) that adjusts hazard scoring coefficients based on ground reports. |
| **Certificate Generation** | ReportLab 5.0.1 | High-fidelity PDF document rendering for official tamper-evident government relocation certificates and SHA-256 integrity hashes. |
| **Schema & Settings** | Pydantic v2 + Pydantic-Settings | Strict data validation, environment variable parsing (.env), and automated request deserialization. |

## 4. Persistence Layer & Spatial Database

Bhoomi Rakshak supports a dual storage strategy accommodating rapid local development and high-concurrency enterprise deployment:

- **Development (SQLite 3 via bhoomi_rakshak.db):** Zero-dependency, embedded file-backed database enabling complete local execution without external service containers.
- **Production (PostgreSQL 15 + PostGIS 3.3):** Enterprise spatial database featuring R-Tree spatial indices (GIST) for sub-millisecond bounding box searches, spatial intersection queries (ST_Intersects), and distance radius filters (ST_DWithin).
- **ORM Layer (SQLAlchemy 2.0):** Modern declarative mapping using SessionLocal connection management and asynchronous driver support via asyncpg.

## 5. Algorithmic Formulations & Mathematical Models

**1. Habitation Priority Scoring Engine**

The priority score objectively determines the urgency of intervention on a normalized 0-100 scale:

**Priority Score = (Hazard Intensity × Exposure × Vulnerability Index) / Normalization Factor**

Where *Hazard Intensity* is derived from intersecting flood/landslide zones, *Exposure* reflects resident population, and *Vulnerability* aggregates socio-economic indicators (% elderly, % disabled, % kutcha housing, low-income bracket).

- **Immediate Phase (Red):** Score ≥ 70 — Urgent structural risk requiring immediate evacuation corridors.
- **Short-Term Phase (Yellow):** Score 40–69 — Escalating risk requiring staged relocation preparation.
- **Medium-Term Phase (Green):** Score < 40 — Safe receiving zone or monitoring zone.

**2. Dijkstra Evacuation Routing with Hazard Impedance Weighting**

Optimal evacuation corridors from danger zones to safe havens are computed using Dijkstra's shortest-path algorithm. Edge weights dynamically incorporate hazard penalties:

**Edge Cost (W) = Geometric Distance (km) × Hazard Impedance Factor**

Impedance Multipliers: **Red Hazard Zone = 1.60x**, **Yellow Hazard Zone = 1.25x**, **Green / Safe Zone = 1.00x**.

This mathematically penalizes routes traversing active disaster corridors, steering evacuees along secure transit arteries.

**3. Relocation Site Carrying Capacity & 8-Factor Suitability**

Proposed relocation parcels are evaluated across eight weighted criteria: (1) Hazard Safety, (2) Carrying Capacity, (3) Road Accessibility, (4) Water Availability, (5) Healthcare Access, (6) Education Access, (7) Livelihood Continuity, and (8) Distance from Origin. Output classifications: *Highly Suitable (≥80)*, *Suitable with Improvements (60-79)*, *Not Suitable (<60)*.

## 6. Artificial Intelligence & External Telemetry Integrations

**Google Gemini Generative AI Integration:** The platform integrates Google's Gemini models (gemini-flash-latest) via an asynchronous conversational endpoint (/api/chatbot). The model is bounded by domain-specific system prompts instructing it in Indian NDMA guidelines, SDRF/NDRF protocols, habitational relocation schemes, and emergency civic reassurance.

**Real-Time Seismic Telemetry (USGS Integration):** live_feeds.py automatically polls the USGS Real-Time Earthquake GeoJSON feed (M2.5+ events) filtered for coordinates covering the Indian subcontinent (Latitude 6°–36°N, Longitude 68°–98°E). Significant tremors dynamically instantiate synthetic hazard boundaries with computed shake intensity.

**Automated Emergency Dispatch (SMTP Services):** Integrated Ethereal SMTP service simulates rapid alert dispatches to district magistrates, block development officers, and emergency responders upon automated trigger of critical red-zone thresholds.

## 7. DevOps, Containerization & Production Verification

| Tool / Service | Configuration & Usage | Production Role |
|---|---|---|
| **Docker & Compose** | Multi-container orchestration via docker-compose.yml | Runs Vite Frontend (5173), FastAPI Backend (8000), and PostGIS (5432) with health-checks. |
| **Oxlint** | Rust-based JavaScript / JSX Linter | Sub-second static analysis ensuring zero syntax regressions and clean JSX component hierarchies. |
| **Print DOM Isolation** | Pure CSS @media print rules | Allows one-click official PDF generation directly from the browser with tamper-evident barcode hashes. |
| **WCAG AAA Compliance** | Contrast ratio > 7:1 across all panels | High readability for emergency field operators in outdoor, high-glare tablet environments. |

## 8. Service Registry & Interactive Access Points

| Service / Resource | Access URL / Endpoint | Protocol & Method |
|---|---|---|
| **Planning Desk & GIS Console** | http://localhost:5173/console | HTTP GET (SPA) |
| **Citizen Resilience Portal** | http://localhost:5173/citizen | HTTP GET (SPA) |
| **FastAPI Swagger Documentation** | http://localhost:8000/docs | HTTP GET (Interactive) |
| **Hazard Overlay Computation** | http://localhost:8000/api/engine/compute | HTTP POST |
| **Live Seismic Data Sync** | http://localhost:8000/api/engine/sync-live | HTTP POST |
| **AI Assistant Chatbot** | http://localhost:8000/api/chatbot | HTTP POST |
| **Official Relocation Certificate** | http://localhost:8000/api/red-zones/{id}/certificate | HTTP GET (PDF Stream) |